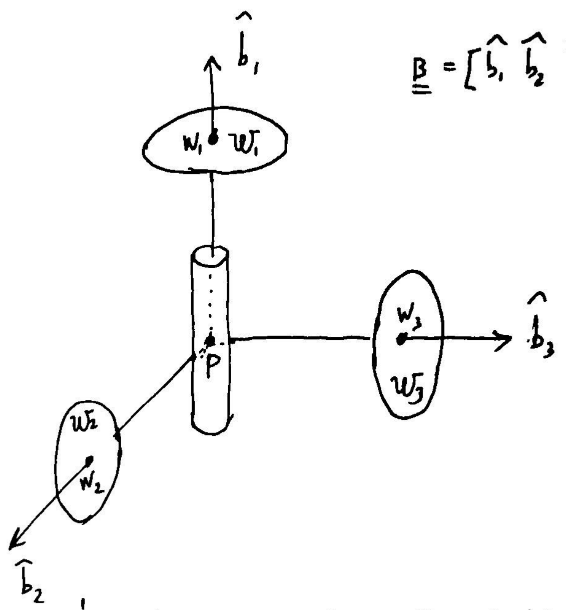
HOMEWORK 1

A spacecraft is modeled as a rigid body $\mathcal{B}$ + damper $\mathcal{D}$ + 3 wheels $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3$ with the geometry depicted in the right figure

$\underline{\underline{B}} = [\hat{b}_1\ \hat{b}_2\ \hat{b}_3]$



Each wheel is a flat disk with mass $m_{W_1} = m_{W_2} = m_{W_3} = 5\ \mathrm{Kg}$ and radius $R = 1\,m$

Distance from P and W_1, W_2, W_3 is $b_W = 2\,m$

The damper has the rest point $R \equiv P$ and mass $m_D = 10\ \mathrm{Kg}$

$\mathcal{B}$ has inertia matrix and mass $m_B = 100\ \mathrm{Kg}$

$$J^{(B)}_{B,P} = \begin{bmatrix} J_1 & 0 & 0 \\ 0 & J_2 & 0 \\ 0 & 0 & J_3 \end{bmatrix},\ \text{with}$$

$J_1 = 350\ \mathrm{Kg\,m^2}$, $J_2 = 300\ \mathrm{Kg\,m^2}$, $J_3 = 400\ \mathrm{Kg\,m^2}$

The damper has constants $K_d = 3.5$ $C_d = 30$

The three wheels are either

(a) free to rotate, i.e. $g_{ai} = 0$ for $\mathcal{W}_i$

(b) rotating with constant angular velocity w_{si} for $\mathcal{W}_i$ (i.e. the initial angular velocity)

(c) subject to viscous damping, i.e.

$$g_{ai} = -K_W w_{si}, \quad \text{where } K_W = 0.1\ \frac{\mathrm{Kg\,m^2}}{\mathrm{sec}}$$

4 cases are considered. For all of them $\underline{v}_p(t_0) = [0 \quad 0 \quad 0]^T$ and $\dot{\xi}(t_0) = 0$, $\xi(t_0) = 0$

Case	$\underline{\omega}^T(t_0)$ (deg/sec)			$\omega_{s1}(t_0)$	$\omega_{s2}(t_0)$ (deg/sec)	$\omega_{s3}(t_0)$	W_1	W_2	W_3	Δt (hours)
1	36	3	3	0	0	0	(a)	(a)	(a)	5
2	36	3	3	1080	0	0	(b)	(b)	(b)	5
3	36	3	3	1080	60	-30	(b)	(c)	(c)	1
4	36	3	3	360	60	-30	(c)	(c)	(c)	1

For each case integrate numerically the dynamics equations and portray the time histories of

(a) components of $\underline{\omega}(t)$

(b) $\omega_{s1}(t)$, $\omega_{s2}(t)$, $\omega_{s3}(t)$

(c) $\xi(t)$ and $\dot{\xi}(t)$

(d) Mechanical energy

__Remark.__ This homework mainly investigate attitude dynamics. Although not very meaningful, the numerical integration yields also $\underline{v}_p(t)$

__Question.__ Explain why the transient behavior in case 3 ends in a much shorter time than in case 2